

AtmosGuard

Is this the sensor, or is this the sky? An anomaly and sensor-health system for an automatic weather station, from temperature, pressure and humidity alone, one station at a time, no neighbours. Smart India Hackathon 2026, problem statement 26073.

VALID

WEATHER: escalate, never delete

SUSPECT: review

FAULT: sensor problem

THE PROBLEM

A stuck barometer and a cyclone look the same on a chart. Cleaning everything deletes the storm; trusting everything feeds bad data into forecasts. Sparse networks have no neighbour to ask.

WHAT IT DOES

- Judges every reading and says why, in words. The raw value is never overwritten.
- Learns each station's own limits; fixed limits alarmed on most clean real data.
- Health score, drift monitor with a stated floor, ticket, service date, estimate with a band.
- Break the sensor on stage and watch it react. ESP32 first layer is one C++ header, tested against the Python.

HOW WE KNOW (NO NEW ALGORITHM CLAIMED)

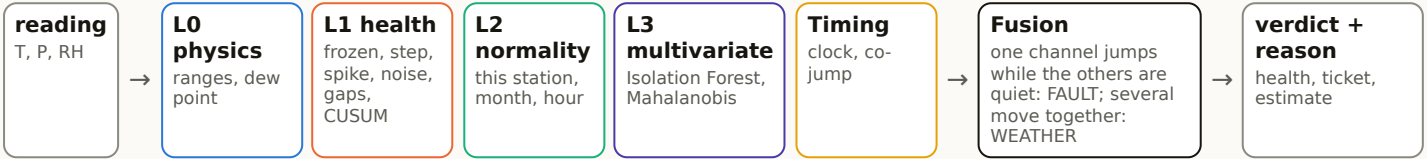
- 62 real stations (NOAA ISD, 2016-24): 31 Indian airports, 12 US northern airports, 12 Australian automatic weather stations, 7 US automated stations reporting every 20 minutes; cyclones, heat and cold waves, outflows.
- Protocol committed first; holdout sealed in time and space, run once, lock file.
- The holdout showed failures: we registered a rule, sealed 12 more stations, ran once: one remedy adopted, one rejected.
- Baselines and an ablation on the same data; NOAA's own flags; one command per result.

RESULTS, SPLIT BY SPLIT (THE NUMBERS ARE NEVER MERGED)

AtmosGuard				Textbook range + step + persistence	
Split	false alarms, clean data	real extreme-weather windows with a FAULT	injected faults detected (mean of 6 types)	false alarms	windows with a FAULT
DEV 6 stations, 2020-21, tuned here	1.9%	0 of 30	95%	5.4%	29 of 30
Holdout in time same 6, 2022-24	2.5%	0 of 38	95%	5.2%	37 of 38
Holdout in space 8 unseen stations	2.9%	3 of 98	88%	6.8%	95 of 98
Fresh stations 12 more, sealed first	2.5%	3 of 139	85%	8.4%	134 of 139
Fresh 2 12 more: 5 airports, 7 AWS	9.3%	4 of 134	91%	3.2%	100 of 134
Fresh 3 12 more: 7 US 20-min, 5 AWS	9.7%	5 of 160	94%	5.1%	157 of 160
Fresh 4 12 more: northern US, freezing winters	2.0%	0 of 180	90%	2.7%	166 of 180

Real records; faults are injected (no labelled real faults exist); "detected" means an alarm the fault itself raised (FAULT or SUSPECT). Full tables, baselines, ablation and intervals: results/REPORT.md.

WHAT ONE READING GOES THROUGH



WHERE IT FAILS (WE SAY IT FIRST)

- Detection is on injected faults; the data are airport records, not IMD AWS records.
- Noise bursts are the weakest class; small drift and constant offsets are invisible from one station.
- 3 of 98 and 3 of 139 and 4 of 134 and 5 of 160 real extreme-weather windows on the unseen stations (holdout, fresh, fresh 2, pipeline as frozen for each) got a FAULT: a 47 % humidity jump in two hours, a 16 C jump across a reporting gap, a 49 % humidity drop in a dry air mass. The step remedy adopted on the last set takes it from 4 windows to 1.
- Simpler detectors beat us on some fault types (spikes; wrong clocks on unseen stations) but miss others and call real weather a fault.
- Most spike, level-shift, noise and clock detections are SUSPECT (review), not FAULT.
- The firmware has not run on hardware.